Szczurek,Santini,Fiacchi - FINAL PROJECT

Aleksandra Szczurek, Lucia Santini, Giada Fiacchi

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This study analyzes the interdependencies between four key variables:

Gold (GLD): SPDR Gold Shares (GLD) is the world's leading gold ETF, providing investors with a simple and liquid way to gain exposure to gold prices without the need to physically hold the metal (Investing.com).

Bitcoin (BTC): BTC/USD is the primary digital currency pair, representing the market price of Bitcoin in U.S. Dollars; it serves as the global benchmark for measuring the cryptocurrency's value and volatility (Investing.com).

Ethereum (ETH): ETH/USD is the currency pair representing the market price of Ether in U.S. Dollars; it serves as the primary benchmark for Ethereum, the leading platform for smart contracts and decentralized finance (Investing.com).

VIX index: The CBOE Volatility Index (VIX) is a real-time market index representing the market's expectation of implied volatility.

Derived from S&P 500 index options, it serves as a primary indicator of systemic risk and investor uncertainty regarding market turbulence over the coming month.

NOTE: The price data for Ethereum were considered starting from April 4, 2016, as in earlier periods trading activity was very limited and trading volumes were not sufficiently high to ensure reliable prices. For this reason, price data for Ethereum prior to this date are not available and could not be included in the analysis; therefore, all data start on April 1, 2016 and end on December 31, 2024.

Using a Latent Markov Model (LMM) (an extension of Hidden Markov Models for continuous data) we aim to identify the underlying market regimes without the use of external covariates.

By leveraging the LMest package in R, we will perform local decoding to map these latent states over time, providing a clear visual representation of market shifts.

```
#import dataset
library(readr)

data <- read_delim("DATITESI.csv",
                   delim = ";",
                   escape_double = FALSE,
                   col_types = cols(
                     P_GLD = col_number(),
```

```
                      P_BTC = col_number(),
                      P_ETH = col_number(),
                      P_VIX = col_number(),
                    ),
                    trim_ws = TRUE)

#convert the "Date" column in date format
data$Date= as.Date(data$Date, format = "%m/%d/%Y")

#reorder the dataset
data <- data[order(data$Date), ]
```

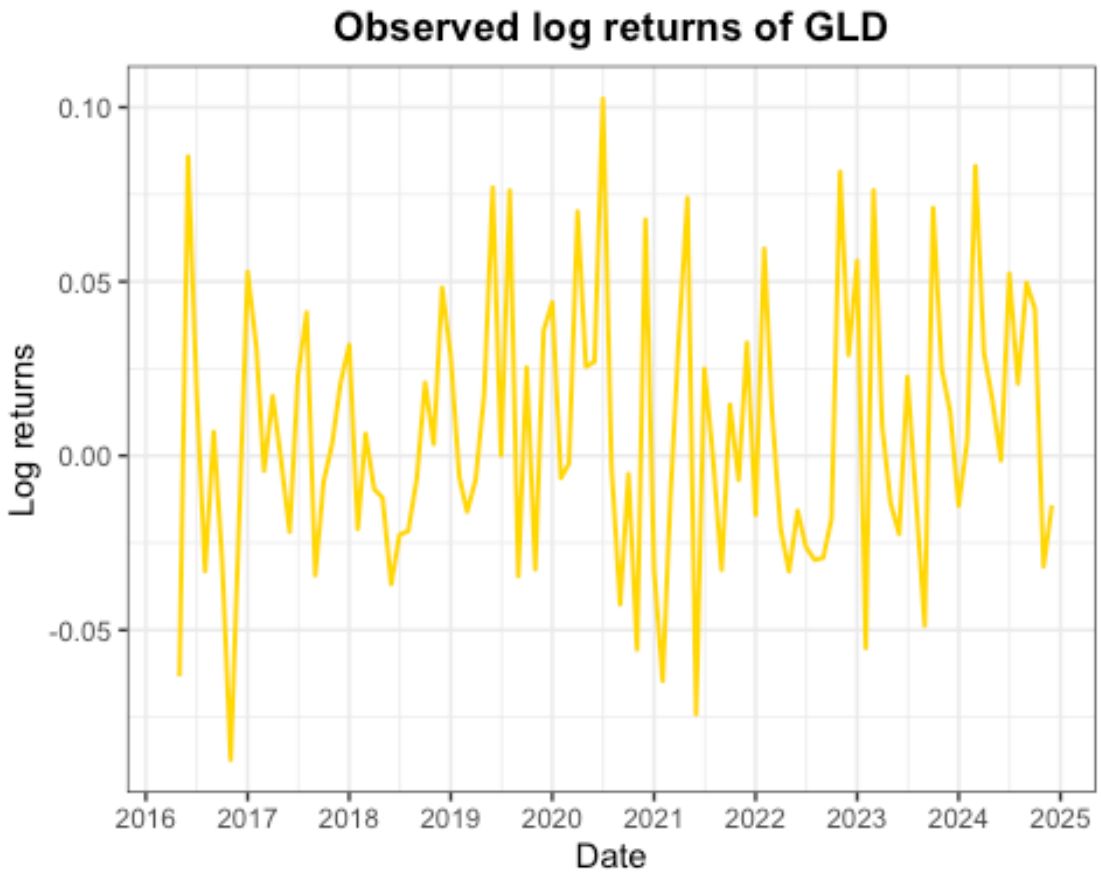
We can try to compute the two models with log-returns: one including VIX and the second excluding it.

We compute the log returns as the difference between: ln(price t) - ln(price t-1)

```
#GLD
data$GLD_log <- c(NA, diff(log(data$P_GLD)))

#plot
library(ggplot2)

ggplot(data, aes(x = Date, y = GLD_log)) +
  geom_line(color = "gold", linewidth = 0.7) +
  ggtitle("Observed log returns of GLD") +
  ylab("Log returns") +
  scale_x_date(date_breaks = "1 year", date_labels = "%Y") +
  theme_bw() +
  theme(
    plot.title = element_text(
      hjust = 0.5,
      face = "bold",
      size = 13
    )
  )
```




```
#descriptive statistics
summary(data$GLD_log)

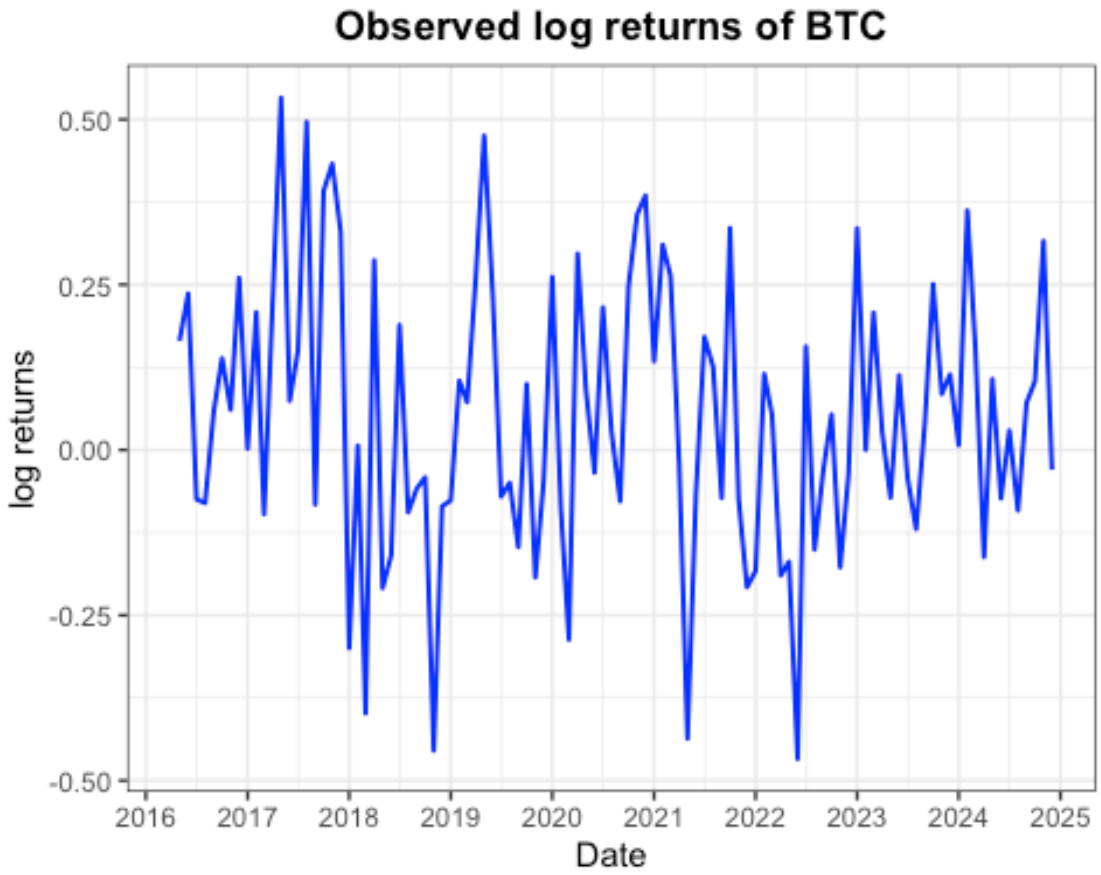
##       Min.    1st Qu.     Median       Mean    3rd Qu.       Max.       NA's
## -0.0872649 -0.0209348 -0.0003454  0.0064617  0.0290282  0.1024705          1
```

The statistics confirm Gold as a stable store of value, showing a steady 0.65% average monthly growth. Its narrow return range (-8.7% to +10.8%) reflects low volatility, while the positive skewness highlights its tendency to gain value during global crises, such as the 2020 pandemic.

```
#BTC
data$BTC_log <- c(NA, diff(log(data$P_BTC)))

#plot
ggplot(data, aes(x = Date, y = BTC_log)) +
  geom_line(color = "blue", linewidth = 0.7) +
  ggtitle("Observed log returns of BTC") +
  ylab("log returns") +
  scale_x_date(date_breaks = "1 year", date_labels = "%Y") +
  theme_bw() +
  theme(
    plot.title = element_text(
      hjust = 0.5,
      face = "bold",
      size = 13
```

```
    )
  )
```




```
#descriptive statistics
summary(data$BTC_log)

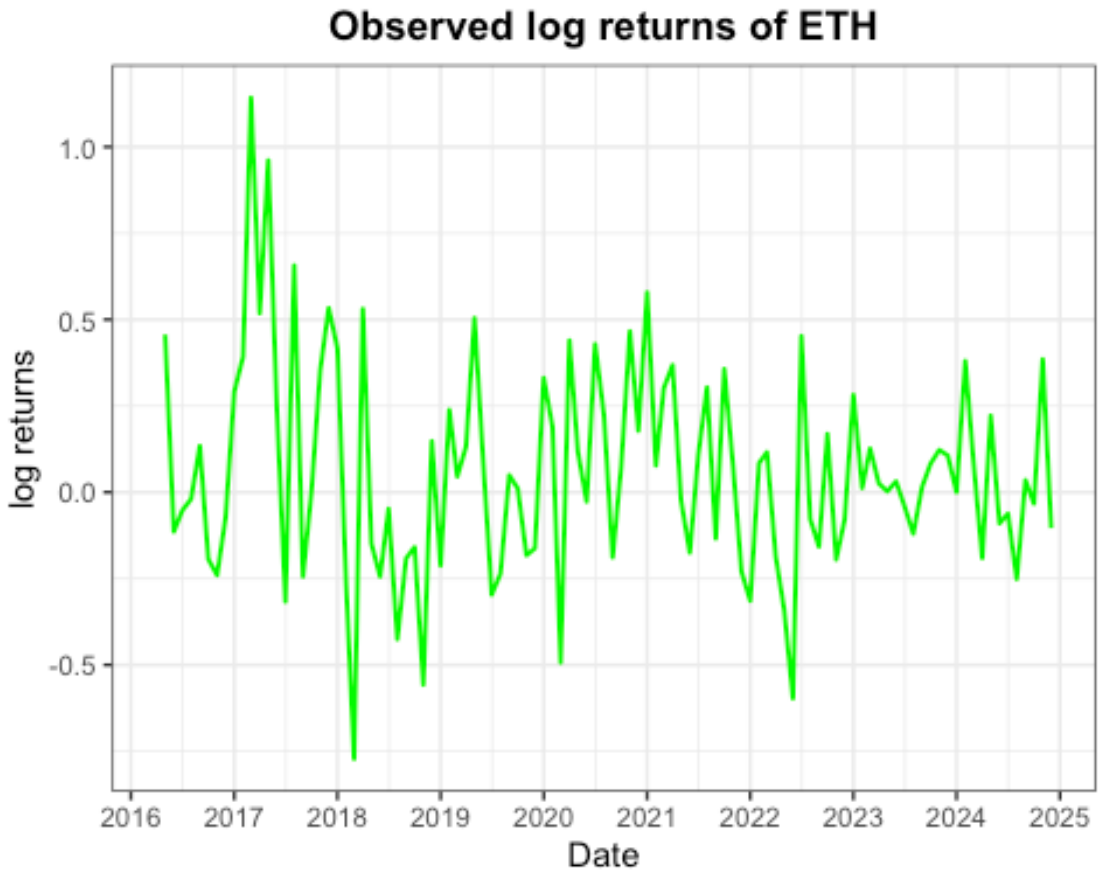
##      Min.   1st Qu.    Median      Mean   3rd Qu.      Max.      NA's
## -0.46720 -0.07653  0.04573  0.05135  0.20799  0.53283         1
```

The statistics define Bitcoin as a high-growth (5.1% monthly mean), high-risk asset with extreme price swings (-46.7% to +53.3%). Its performance is characterized by explosive potential, setting it apart from traditional low-volatility safe havens.

```
#ETH
data$ETH_log <- c(NA, diff(log(data$P_ETH)))

#plot
ggplot(data, aes(x = Date, y = ETH_log)) +
  geom_line(color = "green", linewidth = 0.7) +
  ggtitle("Observed log returns of ETH") +
  ylab("log returns") +
  scale_x_date(date_breaks = "1 year", date_labels = "%Y") +
  theme_bw() +
  theme(
    plot.title = element_text(
```

```
      hjust = 0.5,
      face = "bold",
      size = 13
    )
  )
```




```
#descriptive statistics
summary(data$ETH_log)

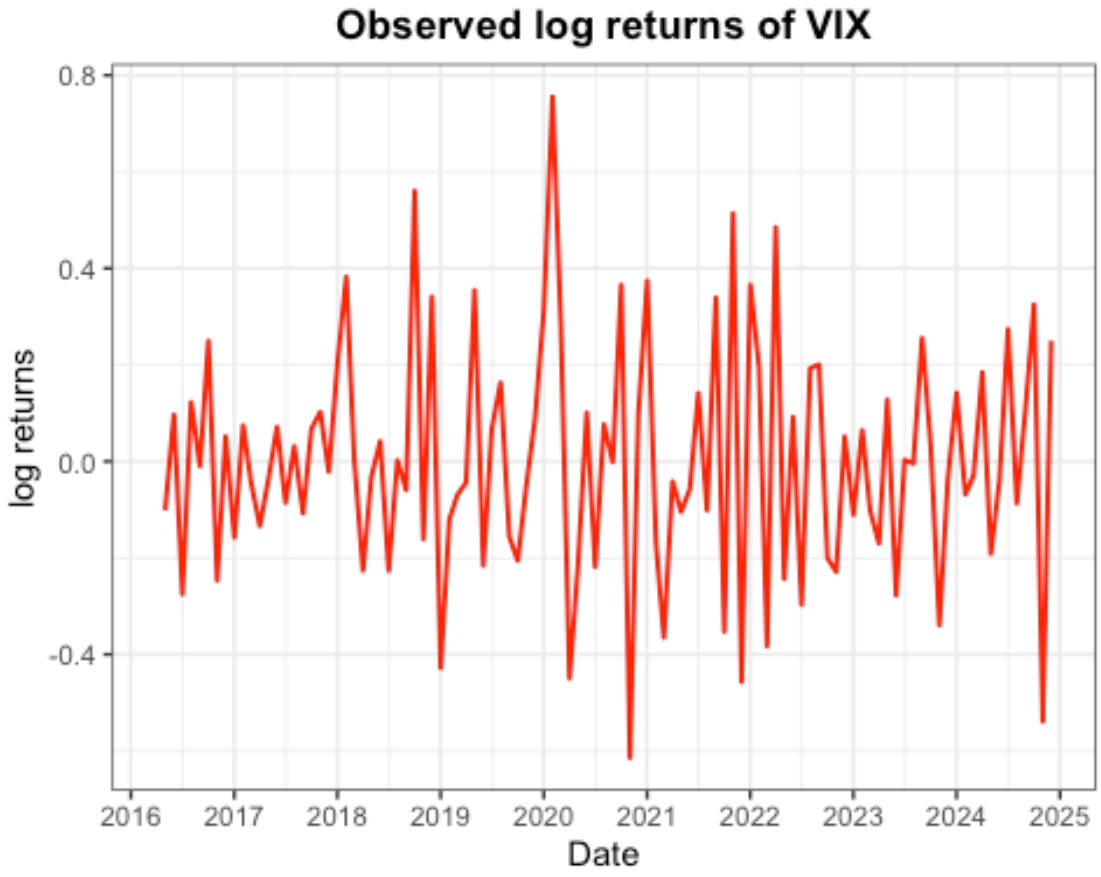
##      Min.   1st Qu.    Median      Mean   3rd Qu.      Max.      NA's
## -0.77191 -0.15973  0.02861  0.05702  0.23439  1.14261         1
```

Ethereum stands as the highest-yielding asset (5.7% monthly mean) but also the most volatile. Its record-breaking range (-77.2% to +114.3%) highlight a profile of extreme market sensitivity, offering the greatest growth potential alongside the highest downside risk in the dataset.

```
#VIX
data$VIX_log <- c(NA, diff(log(data$P_VIX)))

#plot
ggplot(data, aes(x = Date, y = VIX_log)) +
  geom_line(color = "red", linewidth = 0.7) +
  ggtitle("Observed log returns of VIX") +
  ylab("log returns") +
  scale_x_date(date_breaks = "1 year", date_labels = "%Y") +
```

```
  theme_bw() +
  theme(
    plot.title = element_text(
      hjust = 0.5,
      face = "bold",
      size = 13
    )
  )
```




```
#descriptive statistics
summary(data$VIX_log)
```

```
##       Min.    1st Qu.     Median       Mean    3rd Qu.       Max.       NA's
## -0.6142786 -0.1586686 -0.0156202  0.0009609  0.1316447  0.7556434          1
```

VIX_log statistics reveal a mean-reverting behavior characterized by sudden volatility spikes. The significant spread between its minimum (-61.4%) and maximum (+75.6%) confirms its function as a high-sensitivity barometer for global market stress.

```
#dataset to use LMest (LMest uses data frames)
data_fin_log <- data.frame(
  id   = 1,
  time = seq_along(data$BTC_log),
  GLD  = data$GLD_log,
  BTC  = data$BTC_log,
```

```
  ETH  = data$ETH_log,
  VIX  = data$VIX_log
)

#remove na
data_fin_log <- na.omit(data_fin_log)

#we can see a summary
head(round(data_fin_log, 3))
```

```
##   id time    GLD    BTC    ETH    VIX
## 2  1    2 -0.063  0.165  0.457 -0.101
## 3  1    3  0.086  0.236 -0.114  0.097
## 4  1    4  0.020 -0.074 -0.053 -0.275
## 5  1    5 -0.033 -0.080 -0.020  0.123
## 6  1    6  0.007  0.058  0.134 -0.010
## 7  1    7 -0.030  0.139 -0.194  0.250
```

```
#maximum of GLD_log
data_fin_log[which.max(data_fin_log$GLD), ]
```

```
##    id time       GLD       BTC       ETH        VIX
## 52  1   52 0.1024705 0.2155971 0.4290152 -0.2183898
```

Time 52 (2020) highlights a record monthly gain for Gold as a safe haven. During this period, Bitcoin and Ethereum aligned with Gold's upward trend, acting as digital hedges against monetary inflation, while a decreasing VIX signaled a gradual return to market stability.

```
#minimum for BTC_log
data_fin_log[which.min(data_fin_log$BTC), ]
```

```
##    id time        GLD        BTC        ETH        VIX
## 75  1   75 -0.0157836 -0.4672032 -0.5968398 0.09186784
```

The minimum value for BTC_log occurs at time 75, corresponding to 2022. This represents a crucial turning point that marks the beginning of an extremely bearish phase for the crypto market, triggered by a radical shift in the global macroeconomic landscape (the end of the pandemic and the onset of geopolitical conflicts in Ukraine).

```
#minimum for ETH_log
data_fin_log[which.min(data_fin_log$ETH), ]
```

```
##    id time         GLD        BTC        ETH        VIX
## 24  1   24 0.006300113 -0.3983874 -0.7719121 0.00602714
```

ETH_log experiences an extreme idiosyncratic crash by VIX. This is due to a crash caused by a speculative bubble within the crypto market, which did not impact traditional financial markets.

```
#maximum for VIX_log
data_fin_log[which.max(data_fin_log$VIX), ]
```

```
##    id time          GLD         BTC       ETH       VIX
## 47  1   47 -0.006382071 -0.09008591 0.1898341 0.7556434
```

Time 47 corresponds to the end of 2019 and highlights the VIX index's ability to anticipate periods of market turbulence. Although the actual global pandemic peak (March 2020) occurred shortly thereafter, late 2019 was characterized by significant uncertainty and a sudden return of volatility following an unusually calm period.

```
#LMest
library(LMest)

## Package 'LMest' version 3.2.7

mod_log <- lmestCont(responsesFormula = GLD + BTC + ETH + VIX ~ NULL,
                     data = data_fin_log,
                     index = c("id", "time"),
                     k = 3,
                     modBasic = 1,
                     tol = 10^-6)

## Missing data in the dataset dealt with as MAR
## ------------|-------------|-------------|-------------|-------------|----------
---|-------------|
##      mod    |      k      |    start    |     step    |     lk      |    lk-lko
| discrepancy |
## ------------|-------------|-------------|-------------|-------------|----------
---|-------------|
##           1 |           3 |           0 |           0 |     205.262 |
##           1 |           3 |           0 |          10 |     227.944 |    0.2089
75 |   0.0368397 |
##           1 |           3 |           0 |          20 |     236.803 |   0.07906
45 |  0.00544523 |
##           1 |           3 |           0 |          30 |     241.884 | 0.0004377
71 | 8.18527e-05 |
##           1 |           3 |           0 |          31 |     241.885 | 0.0001835
52 | 3.57683e-05 |
## ------------|-------------|-------------|-------------|-------------|----------
---|-------------|

summary(mod_log)

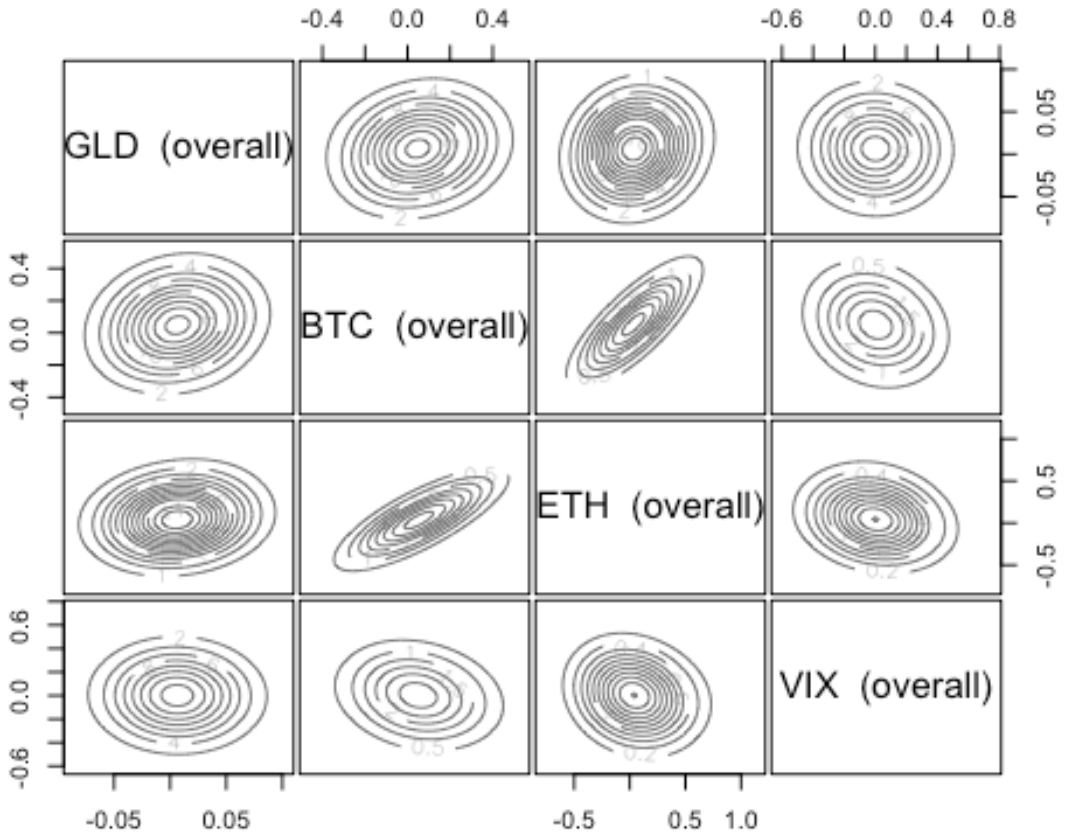
## Call:
## lmestCont(responsesFormula = GLD + BTC + ETH + VIX ~ NULL, data = data_fin_log,
##     index = c("id", "time"), k = 3, modBasic = 1, tol = 10^-6)
##
## Coefficients:
##
## Initial probabilities:
##      est_piv
## [1,]       0
## [2,]       1
## [3,]       0
```

```
##
## Transition probabilities:
##      state
## state      1      2      3
##     1 0.9848 0.0152 0.0000
##     2 0.0000 0.8913 0.1086
##     3 0.6737 0.3263 0.0000
##
##  Mu - Conditional response means:
##      state
## item        1       2       3
##   GLD  0.0087 -0.0027  0.0339
##   BTC  0.0137  0.1927 -0.2769
##   ETH -0.0122  0.1977  0.5116
##   VIX  0.0102 -0.0267  0.0182
##
##  Si - Variance-covariance matrix:
##         [,1]    [,2]    [,3]    [,4]
## [1,]  0.0014  0.0017  0.0016 -0.0002
## [2,]  0.0017  0.0330  0.0387 -0.0077
## [3,]  0.0016  0.0387  0.0793 -0.0107
## [4,] -0.0002 -0.0077 -0.0107  0.0582
```

The Latent Markov Model identifies three distinct market regimes: State 1 represents a period of stability, acting as the standard market condition to which the system reverts following significant shocks. State 2 is characterized by crypto growth, reflecting a confident market that drives cryptocurrency returns upward. State 3 highlights a phase of extreme divergence, where Gold shows decisive growth and Ethereum significantly outperforms Bitcoin despite prevailing market turbulence.

The variance-covariance matrix confirms that while Ethereum remains the most volatile asset, both cryptocurrencies maintain a negative correlation with the VIX, reinforcing their status as risk-on assets within the analyzed timeframe.

```r
#contour plot of the estimated overall density
plot(mod_log, what = "density")
```



The contour plots of the overall joint densities visually confirm the statistical findings of the model. The BTC-ETH relationship shows a tight, positively sloped elliptical shape, indicating a high degree of correlation. In contrast, the nearly circular contours between Gold (GLD) and cryptocurrencies demonstrate a lack of significant correlation, highlighting Gold's role as a diversifying asset. Finally, the slight negative inclination in the VIX-crypto plots supports the identification of digital assets as risk-on instruments, susceptible to spikes in global market volatility.

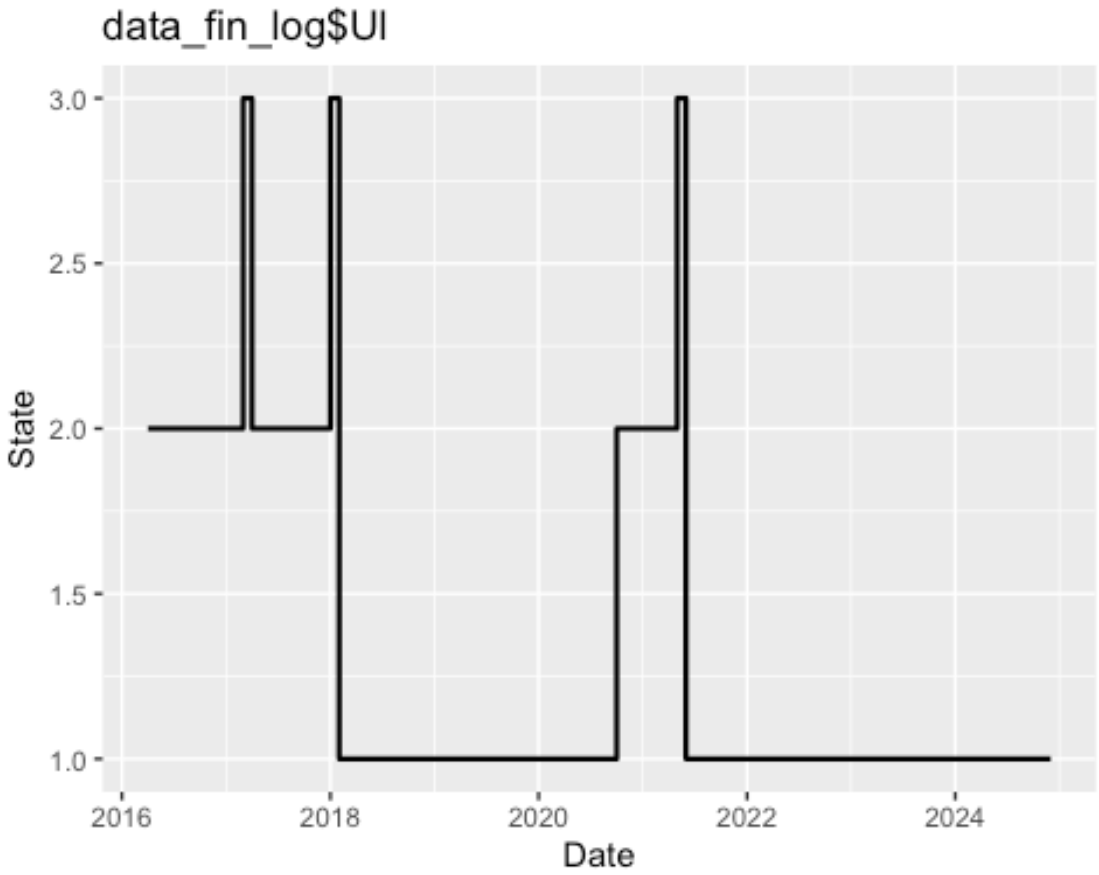
```
#predict the sequence of latent states
deco_log <- lmestDecoding(mod_log)

#local decoding
deco_log$Ul
```

```
##   [1] 2 2 2 2 2 2 2 2 2 2 2 3 2 2 2 2 2 2 2 2 2 3 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1
##  [38] 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 2 2 2 2 2 2 2 3 1 1 1 1 1 1 1 1 1 1 1 1
##  [75] 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1
```

```
#plot
ggplot(data, aes(x = Date, y = deco_log$Ul)) +
  geom_step(color = "black", linewidth = 0.8) +
  labs(
    title = "data_fin_log$Ul",
```

```
    y = "State"
  )
```



The local decoding plot illustrates the temporal progression of the latent states identified by the model. State 2 characterizes the initial market euphoria from 2016 to late 2017. State 1 serves as the primary 'stability' regime, prevailing during market consolidations and bear cycles, showing high persistence after major shocks. Conversely, State 3 acts as a purely impulsive regime, appearing only as brief spikes during extreme market events or asset divergences. This confirms that the analyzed market operates through distinct regime shifts, where periods of high volatility are sharp but short-lived, inevitably reverting to more stable baseline conditions.

```
#### REMOVE VIX VARIABLE ####

#dataset to use LMest (LMest uses data frames)
data_fin_log1 <- data.frame(
  id   = 1,
  time = seq_along(data$GLD_log),
  GLD  = data$GLD_log,
  BTC  = data$BTC_log,
  ETH  = data$ETH_log
  )

data_fin_log1 <- na.omit(data_fin_log1)
```

```r
#we can see a summary
head(round(data_fin_log1, 3))
```

```
##   id time    GLD    BTC    ETH
## 2  1    2 -0.063  0.165  0.457
## 3  1    3  0.086  0.236 -0.114
## 4  1    4  0.020 -0.074 -0.053
## 5  1    5 -0.033 -0.080 -0.020
## 6  1    6  0.007  0.058  0.134
## 7  1    7 -0.030  0.139 -0.194
```

```r
#LMest
mod_log1 <- lmestCont(responsesFormula = GLD + BTC + ETH ~ NULL,
                      data = data_fin_log1,
                      index = c("id", "time"),
                      k = 3,
                      modBasic = 1,
                      tol = 10^-6)
```

```
## Missing data in the dataset dealt with as MAR
## ------------|-------------|-------------|-------------|-------------|----------
---|-------------|
##      mod    |      k      |    start    |     step    |     lk      |    lk-lko
| discrepancy |
## ------------|-------------|-------------|-------------|-------------|----------
---|-------------|
##           1 |           3 |           0 |           0 |      209.24 |
##           1 |           3 |           0 |          10 |     224.756 |    0.7695
32 |    0.083262 |
##           1 |           3 |           0 |          20 |     225.863 |    0.0964
81 |   0.0172959 |
##           1 |           3 |           0 |          30 |     239.344 |    0.1311
82 |    0.027288 |
##           1 |           3 |           0 |          40 |     239.633 |   0.02371
15 |   0.0675889 |
##           1 |           3 |           0 |          50 |     239.852 |  0.004420
34 |   0.0107789 |
##           1 |           3 |           0 |          56 |     239.859 | 0.0002057
52 | 0.000504299 |
## ------------|-------------|-------------|-------------|-------------|----------
---|-------------|
```

```r
summary(mod_log1)
```

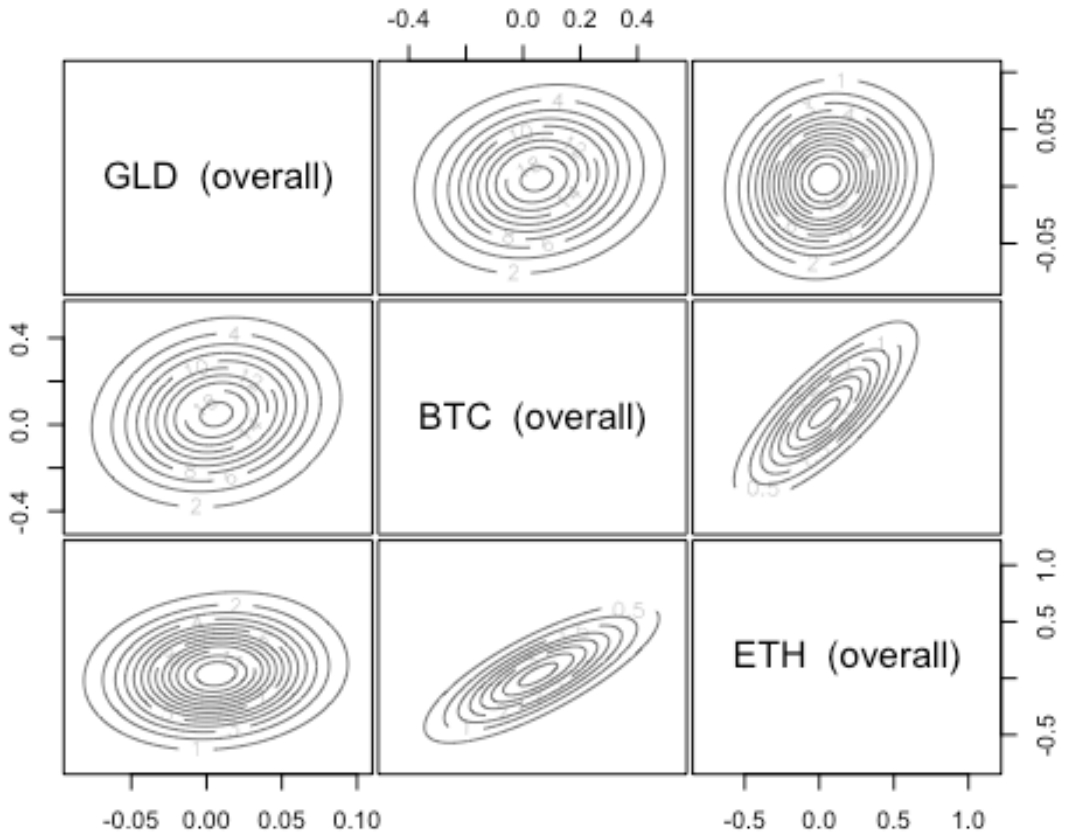
```
## Call:
## lmestCont(responsesFormula = GLD + BTC + ETH ~ NULL, data = data_fin_log1,
##     index = c("id", "time"), k = 3, modBasic = 1, tol = 10^-6)
##
## Coefficients:
##
## Initial probabilities:
```

```
##      est_piv
## [1,]  0.0000
## [2,]  0.9993
## [3,]  0.0007
##
## Transition probabilities:
##      state
## state      1      2      3
##     1 0.9848 0.0152 0.0000
##     2 0.0000 0.8914 0.1086
##     3 0.6737 0.3263 0.0000
##
##  Mu - Conditional response means:
##      state
## item        1       2       3
##   GLD  0.0087 -0.0027  0.0339
##   BTC  0.0137  0.1927 -0.2770
##   ETH -0.0122  0.1978  0.5117
##
##  Si - Variance-covariance matrix:
##        [,1]   [,2]   [,3]
## [1,] 0.0014 0.0017 0.0016
## [2,] 0.0017 0.0330 0.0387
## [3,] 0.0016 0.0387 0.0793
```

The summary of the revised model (excluding the VIX) reinforces the identification of three distinct market regimes. State 1 represents a high-persistence baseline with near-zero returns. State 2 captures a synchronized bull market for digital assets, where Bitcoin and Ethereum exhibit high positive correlation. Conversely, State 3 identifies periods of extreme market divergence: Bitcoin experiences significant losses, while Gold and Ethereum record their highest mean returns.

The variance-covariance matrix (Si) underscores that while digital assets share a strong internal correlation, they remain fundamentally decoupled from Gold. The minimal covariance between GLD and the crypto-pair confirms Gold's consistent role as an independent asset, providing effective diversification regardless of the prevailing latent state of the cryptocurrency market.

```r
#contour plot of the estimated overall density
plot(mod_log1, what = "density")
```



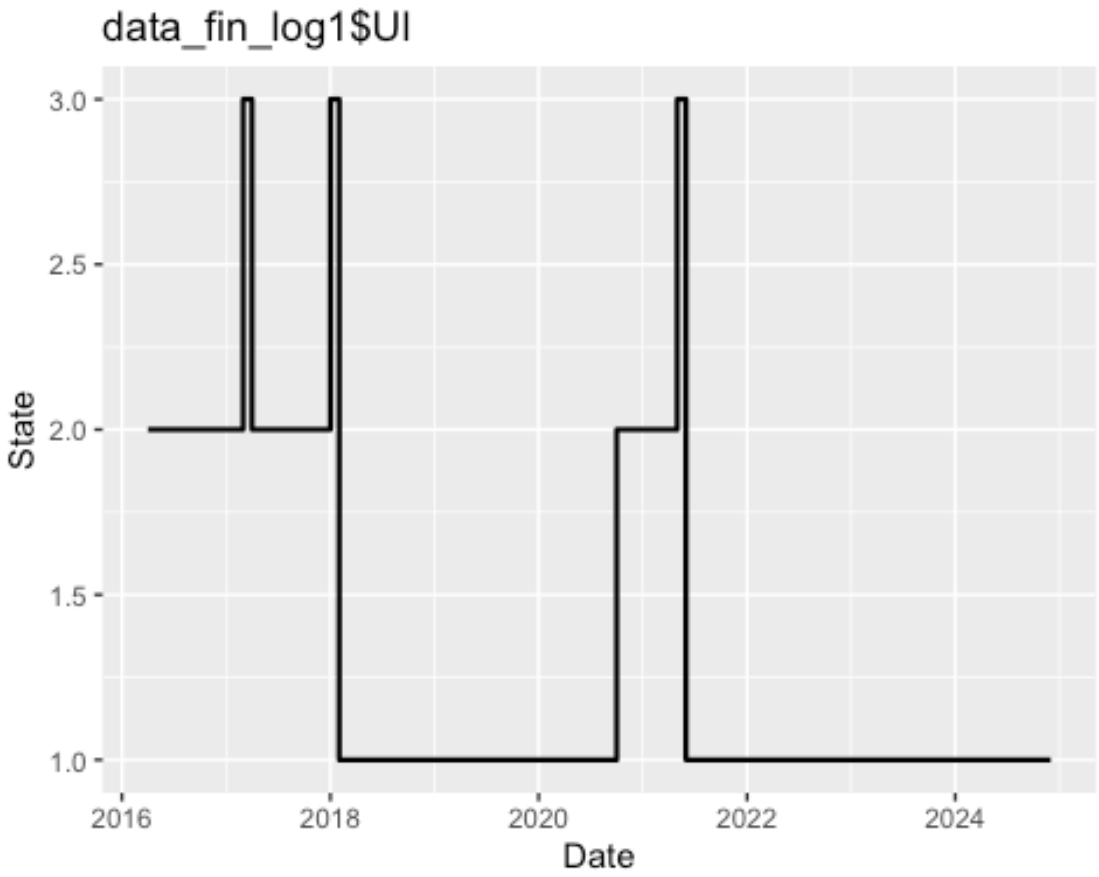
This visual invariance beteween the densities of the two models, confirms that the underlying covariance structure (Si) is stable; the fundamental interdependencies between the assets are intrinsically driven and are not distorted by the inclusion or removal of equity volatility metrics.

```
#predict the sequence of latent states
deco_log1 <- lmestDecoding(mod_log1)

#local decoding
deco_log1$Ul
```

```
##   [1] 2 2 2 2 2 2 2 2 2 2 3 2 2 2 2 2 2 2 2 2 3 1 1 1 1 1 1 1 1 1 1 1 1 1 1
##  [38] 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 2 2 2 2 2 2 2 3 1 1 1 1 1 1 1 1 1 1 1
##  [75] 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1
```

```
#plot
ggplot(data, aes(x = Date, y = deco_log1$Ul)) +
  geom_step(color = "black", linewidth = 0.8) +
  labs(
    title = "data_fin_log1$Ul",
    y = "State"
  )
```



Whether the VIX is included or excluded, the temporal identification of the market regimes remains virtually identical.

The invariance of the decoding sequence suggests that GLD, BTC and ETH provide a sufficiently strong signal to define market regime shifts independently of external equity volatility factors, reinforcing the structural validity of the model.

To understand which model is the best, we can compare them using the Information Criteria (Akaike and Bayesian), tools that allow us to balance the complexity of the model and its goodness of fit.

```
#comparison table
table <- data.frame(
  Model = c("Unrestricted (with VIX)", "Restricted (without VIX)"),
  AIC = c(mod_log$aic, mod_log1$aic),
  BIC = c(mod_log$bic, mod_log1$bic)
); table
```

```
##                       Model       AIC       BIC
## 1  Unrestricted (with VIX) -423.7692 -483.7692
## 2 Restricted (without VIX) -433.7171 -479.7171
```

The comparison table reveals a trade-off between information criteria.

While the AIC favors the restricted model, the Bayesian Information Criterion (BIC) identifies the Unrestricted Model (with VIX) as the best one.

The lower BIC value (-483.76) demonstrates that the explanatory power gained by including equity market volatility outweighs the penalty for the additional parameters.

This suggests that the VIX is a fundamental driver in defining latent market regimes, providing essential context that Gold and digital assets alone cannot fully capture.

In conclusion, this research shows that the relationship between the observed asset is not constant but shifts depending on the different market regimes identified by the states of the Hidden Markov Model.

The resulting model that includes the VIX proved to be the most statistically reliable (thanks to a better BIC score), confirming that fear in the equity markets is a key element in understanding these dynamics.

Moreover, an important aspect that emerged is the solidity of these results: whether or not we consider the traditional stock market, the division of time into these three states remains identical.

This means we have identified a real market structure.